Crispy Fried Chicken Conjecture on Galaxy Dynamics

Carl P. Zimmerman[1]★
[1]Briar Creek Tech, USA

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

Galaxies spin as if they feel more pull than their visible matter can supply. The extra pull appears below one special acceleration, $a_0 \simeq 1.2 \times 10^{-10}$ m s^{-2}. This number is close to $\frac{1}{2}c\sqrt{G\rho_\Lambda}$, where ρ_Λ is the density of the cosmological constant (dark energy). We call the guess $a_0 = \kappa\, c\sqrt{G\rho_\Lambda}$, with κ a pure number, the Crispy Fried Chicken conjecture, and test it with every step of the algebra shown. First, what is κ? On the SPARC galaxy sample three methods give 0.45 ± 0.07, 0.58 ± 0.18 and 0.43 ± 0.08. All agree with $1/2$, and with other simple numbers. The answer depends on how much mass each unit of starlight carries, so $1/2$ is a fitted value, not a derived one. Second, did a_0 change over cosmic time? If it is tied to dark energy, it did not. If it is tied to the expansion rate, it was 3.8 times larger at redshift $z = 2.5$. Dark-matter (ΛCDM) haloes predict about 1.6 times. Working out a_0 from one galaxy magnifies the error in its visible mass by 4 or more at the accelerations of today's distant galaxies, so they cannot decide. About eight magnified (lensed) discs at $z \simeq 2.5$ would give 20:1 odds if their mass errors were independent. If the errors share a common calibration error of 0.06 dex (15 per cent), 34 are needed, and above that the sample average alone can never reach 20:1.

Key words: gravitation – galaxies: kinematics and dynamics – galaxies: high-redshift – cosmological parameters – dark energy

1 INTRODUCTION

Stars and gas in a spinning galaxy are pulled toward its centre with acceleration $g_{\rm obs} = V^2/R$, where V is their speed at distance R from the centre. The visible stars and gas alone would pull less; call that smaller value $g_{\rm bar}$ ("bar" is for baryons, meaning ordinary matter). Across thousands of measurements $g_{\rm obs}$ follows one curve of $g_{\rm bar}$ (McGaugh et al. 2016; Lelli et al. 2017). The curve has one built-in constant, an acceleration $a_0 \simeq 1.2\times10^{-10}$ m s^{-2}. Where $g_{\rm bar} \gg a_0$, $g_{\rm obs} = g_{\rm bar}$. Where $g_{\rm bar} \ll a_0$, $g_{\rm obs} \simeq \sqrt{g_{\rm bar} a_0}$. Milgrom (1983) proposed this as a modified law of motion (MOND) and noted that a_0 is about cH_0, where c is the speed of light and H_0 is how fast the universe expands today.

That coincidence can be read three ways. (i) a_0 follows the expansion rate $H(z)$, where redshift z measures how far back in time we look; then a_0 was larger in the past. (ii) a_0 follows the cosmological constant, which never changes; then a_0 is constant. (iii) There is no basic a_0, and in ΛCDM the curve emerges from dark-matter haloes (Navarro et al. 2017), whose pull also grows with redshift. Today (i) and (ii) differ by only 21 per cent, too little to measure. At $z = 2.5$ they differ by a factor 3.8.

We test reading (ii), written as

$$a_0 = \kappa\, c\sqrt{G\rho_\Lambda}. \tag{1}$$

Here G is Newton's constant, ρ_Λ is the mass density of the cosmological constant and κ is a pure number. We call this the Crispy Fried Chicken conjecture; the name is the author's nickname for it and carries no physics. We size differences in *dex*: 0.3 dex is a factor 2, 1 dex a factor 10, and 0.06 dex is 15 per cent. We ask what κ is (Section 3), and what it would take to test whether a_0 changes with redshift (Section 4). The idea is old (Milgrom 1999, 2020), and Limbach et al. (2008) ran a redshift test up to $z = 1.2$; we claim no priority. We do not derive κ.

2 THE CONJECTURE, ONE STEP AT A TIME

Why the form is forced. Suppose an acceleration is built from c, G and a density ρ as $a = c^\alpha G^\beta \rho^\gamma$. In kilograms (kg), metres (m) and seconds (s), c is m s^{-1}, G is m^3 kg^{-1} s^{-2}, ρ is kg m^{-3} and an acceleration is m s^{-2}. The power of each unit must match on both sides:

$$\begin{aligned} \text{kg:}\quad & -\beta + \gamma = 0, \\ \text{m:}\quad & \alpha + 3\beta - 3\gamma = 1, \\ \text{s:}\quad & -\alpha - 2\beta = -2. \end{aligned} \tag{2}$$

The first line gives $\gamma = \beta$. Putting that into the second gives $\alpha = 1$. Putting $\alpha = 1$ into the third gives $\beta = 1/2$. So $a = \kappa c\sqrt{G\rho}$ is the only possible form, and units cannot fix κ. Choosing $\rho = \rho_\Lambda$ is the conjecture.

The density of dark energy. For a flat universe $H^2 = 8\pi G\rho/3$. Today the total density is the critical density, and the part of it that is the cosmological constant is

$$\rho_{\rm crit} = \frac{3H_0^2}{8\pi G}, \qquad \rho_\Lambda = \Omega_\Lambda\, \rho_{\rm crit}. \tag{3}$$

Unlike matter, ρ_Λ does not thin out as the universe expands. We use the Planck Collaboration et al. (2020) values $H_0 = 67.4$ km s^{-1} Mpc^{-1} and $\Omega_\Lambda = 0.685$, with 1 Mpc $= 3.08568 \times 10^{22}$ m. Then

★ E-mail: address supplied at submission

© 2026 The Authors

equation (1) works out step by step as

$$\begin{aligned} H_0 &= \frac{67.4\times10^3}{3.08568\times10^{22}} = 2.184\times10^{-18}\ \mathrm{s^{-1}},\\ \rho_{\rm crit} &= \frac{3H_0^2}{8\pi G} = 8.533\times10^{-27}\ \mathrm{kg\,m^{-3}},\\ \rho_\Lambda &= 0.685\,\rho_{\rm crit} = 5.845\times10^{-27}\ \mathrm{kg\,m^{-3}},\\ \sqrt{G\rho_\Lambda} &= 6.246\times10^{-19}\ \mathrm{s^{-1}},\\ c\sqrt{G\rho_\Lambda} &= 1.872\times10^{-10}\ \mathrm{m\,s^{-2}}. \end{aligned} \tag{4}$$

For $\kappa = 1/2$ this gives $a_0 = 9.36\times10^{-11}$ m s^{-2}. Using $\rho_{\rm crit}$ instead of ρ_Λ makes it larger by $\Omega_\Lambda^{-1/2} = 1.208$, which is 1.13×10^{-10} m s^{-2}. Both lie inside the stated uncertainty of the usual value, $(1.20\pm0.02\pm0.24)\times10^{-10}$ m s^{-2} (McGaugh et al. 2016).

3 MEASURING κ

The curve. We use the SPARC database (Lelli et al. 2016): 175 disc galaxies, of which we keep 2803 points in 164 galaxies with velocity errors below 10 per cent. The curve, in the form of McGaugh et al. (2016), is

$$g_{\rm obs} = \nu(y)\,g_{\rm bar},\quad \nu(y) = \frac{1}{1-e^{-\sqrt{y}}},\quad y \equiv \frac{g_{\rm bar}}{a_0}. \tag{5}$$

For $y \gg 1$, $e^{-\sqrt{y}} \to 0$, so $\nu \to 1$ and $g_{\rm obs} = g_{\rm bar}$. For $y \ll 1$, write $s = \sqrt{y}$ and expand $1-e^{-s} = s - s^2/2 + \cdots = s(1 - s/2 + \dots)$. Then $\nu = (1/s)(1 + s/2 + \dots)$ and

$$g_{\rm obs} = \frac{g_{\rm bar}}{\sqrt{y}}\left(1+\tfrac{1}{2}\sqrt{y}\right) = \sqrt{g_{\rm bar}a_0}\left(1+\tfrac{1}{2}\sqrt{y}\right), \tag{6}$$

because $g_{\rm bar}/\sqrt{y} = g_{\rm bar}\sqrt{a_0/g_{\rm bar}} = \sqrt{g_{\rm bar}a_0}$. The leading term says $a_0 = g_{\rm obs}^2/g_{\rm bar}$.

Why κ depends on the stars. The pull $g_{\rm obs} = V^2/R$ is measured. But $g_{\rm bar}$ is computed from the light of the stars and the gas, and the stellar part is proportional to Υ, the mass each unit of starlight carries, which comes from stellar models and is not measured. Where stars dominate, $g_{\rm bar} \propto \Upsilon$, so $a_0 = g_{\rm obs}^2/g_{\rm bar} \propto 1/\Upsilon$. Fitting equation (5) to SPARC gives $a_0 = 1.05$, 0.86 and 0.71×10^{-10} m s^{-2}, that is $\kappa = 0.560$, 0.457 and 0.381, for $\Upsilon = 0.5$, 0.6 and 0.7. Going from $\Upsilon = 0.5$ to 0.6 multiplies κ by $0.457/0.560 = 0.82$, close to $0.5/0.6 = 0.83$. Stellar models give $\Upsilon = 0.5$–0.7 (Meidt et al. 2014), and $\kappa = 1/2$ needs $\Upsilon = 0.555$. The scatter hardly changes (0.143–0.145 dex), so the data do not choose Υ. One more choice matters: SPARC places 97 galaxies at distances computed with $H_0 = 73$, while ρ_Λ uses 67.4. We rescale those distances to 67.4 so both sides of equation (1) use one H_0.

Three methods. To loosen the grip of Υ we use three methods that depend on it differently. A: the rule that a galaxy's flat rotation speed V_f obeys $V_f^4 \simeq GM_b a_0$ (M_b is the visible mass), corrected for galaxies that are not infinitely deep, for 24 deep galaxies. B: only the shape of the curve, where it bends, with a free vertical shift so that a wrong distance scale cannot matter. C: all 3391 points, letting each galaxy choose its own Υ. They give $\kappa = 0.450\pm0.073$, 0.58 ± 0.18 and 0.43 ± 0.08 (total errors). Each agrees with $\kappa = 1/2$ (0.7, 0.5 and 0.9σ away) and with Milgrom's $a_0 = cH_\Lambda/2\pi$, which has $\kappa = 0.461$ (0.1, 0.7 and 0.4σ), where $H_\Lambda = H_0\sqrt{\Omega_\Lambda}$. Against this, the study of Desmond (2023), which also fits distances and inclinations, gives $\kappa = 0.636\pm0.053$, 2.6σ above 1/2. Changing the H_0 choice moves our three values by 0.05 to 0.10.

κ cannot be separated from H_0. Equation (3) gives $\sqrt{G\rho_\Lambda} = H_0\sqrt{3\Omega_\Lambda/8\pi}$, so $a_0 = \kappa\, cH_0\sqrt{3\Omega_\Lambda/8\pi}$: only the product κH_0 matters. The two current values $H_0 = 67.4$ and 73.04 (Riess et al. 2022) have the ratio 0.923, and $0.4607/0.500 = 0.921$, so $(\kappa, H_0) = (0.500, 67.4)$ and $(0.461, 73.04)$ predict the same a_0 to 0.2 per cent. We use $\kappa = 1/2$ as a working value because it is the simplest number that fits, and for no deeper reason.

4 DID A_0 CHANGE OVER COSMIC TIME?

4.1 Three predictions

Let $\Delta(z) \equiv \log_{10}[a_0(z)/a_0(0)]$, the change since today in dex. *(a) Tied to dark energy:* ρ_Λ does not change, so $\Delta = 0$. Plain MOND predicts the same; the dark-energy link adds a reason for it. *(b) Tied to the critical density:* with $\rho_{\rm crit}(z) = 3H^2(z)/8\pi G$ we get $a_0 \propto H(z)$ and $\Delta = \log_{10} E(z)$, where $E(z) = H(z)/H_0 = \sqrt{\Omega_m(1+z)^3+\Omega_\Lambda}$ and $\Omega_m = 0.315$ is the matter share. At $z = 2.5$,

$$E^2 = 0.315\times3.5^3 + 0.685 = 14.19, \tag{7}$$

so $E = 3.77$ and $\Delta = \log_{10} 3.77 = +0.58$. *(c) Dark-matter haloes (ΛCDM):* the pull at the centre of a halo of mass M and concentration c_{200} is $g_{\rm max} \propto (GM)^{1/3}H^{4/3}c_{200}^2/f(c_{200})$, with $f(x) = \ln(1+x) - x/(1+x)$ (derivation: Appendix of Zimmerman 2026). At fixed M the factor $H^{4/3}$ gives $\frac{4}{3}\log_{10} E = +0.77$ at $z = 2.5$, because haloes were denser; but older haloes were less concentrated, which subtracts 0.44 (Dutton & Macciò 2014). The sum is +0.33 for 10^{12} M$_\odot$ and +0.22 for the lighter haloes of the discs of Section 4.4; across halo mass and model (Duffy et al. 2008), +0.17 to +0.56.

4.2 Why one galaxy tells you so little

Take one galaxy at one radius. Solve $\nu(y) = g_{\rm obs}/g_{\rm bar}$ for y, and set $\hat{a}_0 = g_{\rm bar}/y$. How do measurement errors reach $\hat{a}_0$?

(i) Take the logarithm of $g_{\rm obs} = \nu(y)g_{\rm bar}$: $\ln\nu(y) = \ln g_{\rm obs} - \ln g_{\rm bar}$.

(ii) Make small changes. With $n \equiv \mathrm{d}\ln\nu/\mathrm{d}\ln y$: $n\,\mathrm{d}\ln y = \mathrm{d}\ln g_{\rm obs} - \mathrm{d}\ln g_{\rm bar}$.

(iii) From $\hat{a}_0 = g_{\rm bar}/y$: $\mathrm{d}\ln\hat{a}_0 = \mathrm{d}\ln g_{\rm bar} - \mathrm{d}\ln y$.

(iv) Put $\mathrm{d}\ln y$ from step 2 into step 3:

$$\mathrm{d}\ln\hat{a}_0 = -\frac{1}{n}\,\mathrm{d}\ln g_{\rm obs} + \left(1+\frac{1}{n}\right)\mathrm{d}\ln g_{\rm bar}. \tag{8}$$

(v) For equation (5), $\ln\nu = -\ln(1-e^{-s})$ with $s = \sqrt{y}$, so $\mathrm{d}\ln\nu/\mathrm{d}s = -1/(e^s-1)$. With $\mathrm{d}\ln y = 2\,\mathrm{d}s/s$,

$$n = -\frac{\sqrt{y}/2}{e^{\sqrt{y}}-1}. \tag{9}$$

Here $n < 0$, so define $A_{\rm obs} = 1/|n|$ and $A_{\rm bar} = A_{\rm obs} - 1$. A fractional error ϵ in $g_{\rm obs}$ becomes $A_{\rm obs}\epsilon$ in $\hat{a}_0$, and one in $g_{\rm bar}$ becomes $A_{\rm bar}\epsilon$. For small y, $n \to -1/2$, so $A_{\rm obs} = 2$ and $A_{\rm bar} = 1$. Both grow with y (Fig. 1b): $A_{\rm bar} = 1.1$ at $y = 0.01$, 1.7 at $y = 0.3$, 3.8 at $y = 2.4$ and 8.1 at $y = 6.4$. So errors are least harmful at low acceleration.

4.3 What existing data say

The largest uniform sample of distant galaxies, RC100 (Nestor Shachar et al. 2023, 100 discs, $0.6 < z < 2.5$), takes visible masses from starlight plus gas estimated from scaling relations (Tacconi et al. 2018). Its median is $y = 2.4$ and its central 68 per cent lies in $0.9 < y < 7.2$. At the median $A_{\rm bar} = 3.8$, so a typical 0.2 dex error in

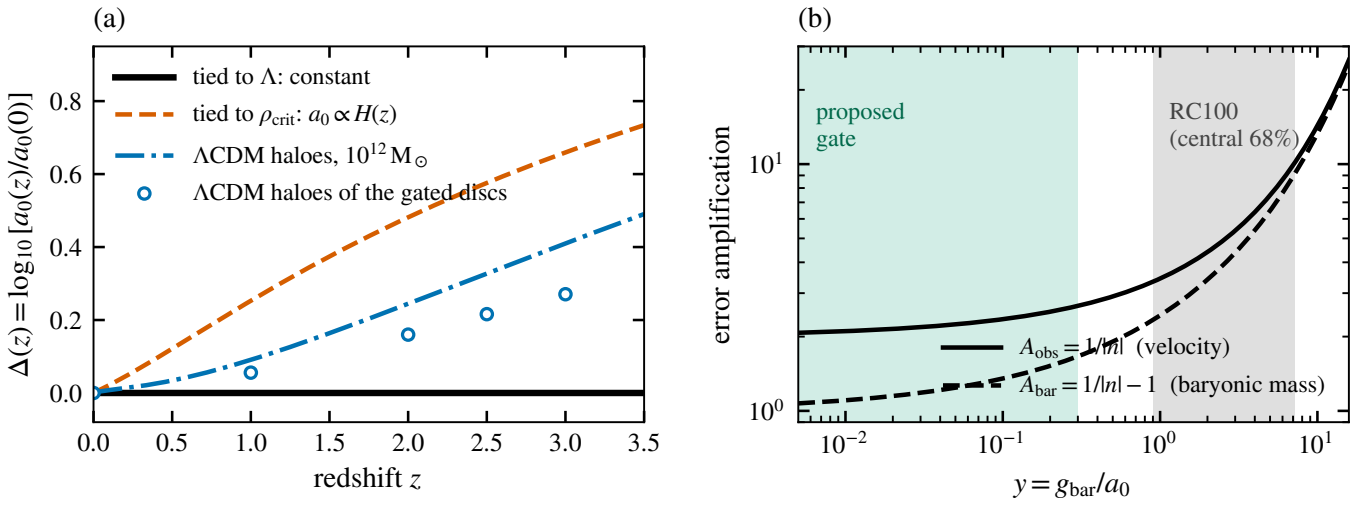


Figure 1. (a) The three predictions of Section 4.1. The circles are the ΛCDM rise for the haloes of the discs selected by the gate of Section 4.4. (b) How many times an error in the observed pull ($A_{\rm obs}$) and in the visible mass ($A_{\rm bar}$) is magnified when a_0 is worked out from one galaxy (equation 8). Green: the proposed gate, $y < 0.3$. Grey: the central 68 per cent of the RC100 sample.

visible mass becomes 0.76 dex in a_0, larger than any gap in Fig. 1a. For 41 of the 100 discs the visible matter alone already supplies the observed pull, so no a_0 can be worked out (27 per cent of discs at $z < 1$, 51 per cent at $z > 2$). RC100 therefore measures how well visible masses are known, not a_0. The rise at $z \simeq 1$ reported by Ciocan et al. (2026) appears when disc masses are fitted together with a dark halo and not when they come from starlight (Zimmerman 2026, Sect. 4.3), so we count it neither way. No existing sample decides.

4.4 What would decide

The gate. Use only discs whose visible pull at their outermost radius is $g_{\rm bar} < 0.3a_0$, where $A_{\rm obs} \le 2.7$ and $A_{\rm bar} \le 1.7$. With $g_{\rm bar} = \Gamma G M_b/R^2$ and a shape factor $\Gamma \simeq 1.2$, this limit on the visible mass M_b is

$$M_b < \frac{0.3\,a_0 R^2}{\Gamma G} = \frac{0.3\,(9.36\times10^{-11})(3.086\times10^{19})^2}{1.2\,(6.674\times10^{-11})}\ \mathrm{kg} = 1.7\times10^8\,\mathrm{M}_\odot\,(R/\mathrm{kpc})^2, \tag{10}$$

using 1 kpc = 3.086×10^{19} m and $\mathrm{M}_\odot = 1.989\times10^{30}$ kg. At $z \simeq 2.5$ such discs can be resolved only when a foreground galaxy magnifies them (gravitational lensing), and their gas mass must be measured, not guessed.

How many galaxies? Each galaxy gives one value Δ_i. We compare two predictions, 0 and Δ. For a Gaussian error σ the log of the likelihood ratio (how much better one prediction fits) is $[x^2 - (x-\Delta)^2]/2\sigma^2 = (2x\Delta - \Delta^2)/2\sigma^2$. If Δ is true, the average of x is Δ, so the expected value is $\langle \ln B\rangle = \Delta^2/2\sigma^2$, and for N independent galaxies $N\Delta^2/2\sigma^2$. Odds of 20:1 need $N\Delta^2/2\sigma^2 = \ln 20$, that is

$$\frac{\sigma}{\sqrt{N}} \le \frac{\Delta}{\sqrt{2\ln 20}} = \frac{\Delta}{2.45}. \tag{11}$$

So the sample mean must be known to $0.216/2.45 = 0.088$ dex to test against the ΛCDM rise of the gated discs, and to $0.576/2.45 = 0.24$ dex against $H(z)$. A fuller count notes that the two predictions also differ in spread: $s_0^2 = \sigma_{\rm m}^2 + \sigma_{\rm int}^2$ if a_0 is constant, and $s_1^2 = s_0^2 + \sigma_{\rm h}^2$ for haloes. Here $\sigma_{\rm m}$ is the measurement error, $\sigma_{\rm int} = 0.034 A_{\rm obs}(0.3) = 0.09$ dex is the natural scatter of the curve, and $\sigma_{\rm h} = 0.135$ dex is the scatter between haloes. If a_0 is constant, each galaxy then adds

$$\ln\frac{s_1}{s_0} - \frac{1}{2} + \frac{s_0^2 + \Delta^2}{2s_1^2} \tag{12}$$

to $\langle \ln B\rangle$ (the less favourable case; if the haloes are right it is larger, 0.513 below). For $\sigma_{\rm m} = 0.20$ dex, $s_0 = 0.2195$ and $s_1 = 0.2575$, so each galaxy adds 0.375: eight galaxies give $e^{8\times0.375} = 20:1$ and seven give 14 : 1. Expected odds are not the chance of reaching them. By simulation that chance is 0.53 for eight galaxies, and 20 are needed for 0.9.

A shared calibration error does not average down. Part of the mass error is the same for every galaxy (the conversion from CO light to gas mass, the stellar-mass zero point). A shared error δ_c (in dex) adds the same $\sigma_C = A_{\rm bar}\delta_c$ to every Δ_i, so the variance of the sample mean is $s^2/N + \sigma_C^2$, which stops falling at σ_C^2. Using σ_C in place of $\sigma/\sqrt{N}$ above, odds of 20:1 from the means need $\delta_c \le \Delta/(2.45\,A_{\rm bar})$. At $y = 0.2$, $A_{\rm bar} = 1.52$, so

$$\delta_c \le \frac{0.216}{2.45\times1.52} = 0.058\ \text{dex (haloes)},\qquad \delta_c \le \frac{0.576}{2.45\times1.52} = 0.155\ \text{dex}\ (H(z)). \tag{13}$$

Beyond these limits the sample average alone can never reach 20:1, however many galaxies are used. (The spread of the sample also differs between the two predictions and adds some information: at $\delta_c = 0.07$ dex, 52 galaxies would give 20:1.) Below the limits the number needed rises, from eight at $\delta_c = 0$ to 14, 20 and 34 at $\delta_c = 0.04$, 0.05 and 0.06 dex. Deep galaxies cannot fit this calibration themselves: if every mass is multiplied by f, then $g_{\rm bar} \to f g_{\rm bar}$ and $g_{\rm obs}^2 = g_{\rm bar}a_0$ becomes $g_{\rm obs}^2 = g_{\rm bar}(f a_0)$, so they measure only the product $f a_0$. The calibration must come from outside the rotation curves, in practice from gas masses measured galaxy by galaxy. The test is the mean of the Δ_i over the gated galaxies. Near 0 it supports a constant a_0; near +0.22 or above, an emergent or $H(z)$-tied scale; and if it fits none of them, it counts against all.

5 LIMITS, AND WHAT IS MISSING

κ is not derived: it is fitted, and 0.461 fits as well. *A number is not a mechanism:* equation (1) fixes a value, not whether galaxies need modified gravity, modified inertia or dark matter, and it does not remove the need for the cold matter measured by the microwave background. If dark energy evolves (DESI Collaboration et al. 2025), $a_0 \propto \sqrt{\rho_{\rm DE}}$ would change by only about -0.1 dex at $z = 2.5$. In short, $1/2$ fits SPARC but is not derived, and the missing input for a decision is gas mass measured galaxy by galaxy at $z \simeq 2.5$.

ACKNOWLEDGEMENTS

Generative AI tools were used. Large language models assisted with code, literature checks, transcribing published data and drafting the text: Anthropic's Claude (mainly via Claude Code, which also rewrote this short version), and models from OpenAI, DeepSeek, Zhipu AI (GLM), Alibaba (Qwen), Google (Gemini, Gemma), Tencent (Hunyuan), Moonshot AI (Kimi) and xAI (Grok). The author checked every result, has reviewed the full content and takes sole responsibility for it. We used the SPARC database (Lelli et al. 2016) and the table of Nestor Shachar et al. (2023). No funding; no conflict of interest.

DATA AVAILABILITY

Every number is produced by `short_numbers.py` (figure: `make_figure.py`) in `https://github.com/carlzimmerman/zimmerman-formula`, directory `qwen_claude_field_theory/papers_2026/mnras_submission_2026_v4_short`. Measured quantities are read from the committed outputs of the longer analysis (`mnras_submission_2026_v3`, tag `mnras-v3.4`; Zimmerman 2026). SPARC: `http://astroweb.cwru.edu/SPARC/`.

REFERENCES

Ciocan B. I., Bouché N. F., Fensch J., Krajnović D., Freundlich J., Desmond H., Famaey B., Techi R., 2026, A&A, 709, L16
DESI Collaboration et al., 2025, Phys. Rev. D, 112, 083515
Desmond H., 2023, MNRAS, 526, 3342
Duffy A. R., Schaye J., Kay S. T., Dalla Vecchia C., 2008, MNRAS, 390, L64
Dutton A. A., Macciò A. V., 2014, MNRAS, 441, 3359
Lelli F., McGaugh S. S., Schombert J. M., 2016, AJ, 152, 157
Lelli F., McGaugh S. S., Schombert J. M., Pawlowski M. S., 2017, ApJ, 836, 152
Limbach C. M., Psaltis D., Özel F., 2008, The redshift evolution of the Tully-Fisher relation as a test of modified gravity, preprint (arXiv:0809.2790)
McGaugh S. S., Lelli F., Schombert J. M., 2016, Phys. Rev. Lett., 117, 201101
Meidt S. E., et al., 2014, ApJ, 788, 144
Milgrom M., 1983, ApJ, 270, 365
Milgrom M., 1999, Physics Letters A, 253, 273
Milgrom M., 2020, The a_0 – cosmology connection in MOND, preprint (arXiv:2001.09729)
Navarro J. F., Benítez-Llambay A., Fattahi A., Frenk C. S., Ludlow A. D., Oman K. A., Schaller M., Theuns T., 2017, MNRAS, 471, 1841
Nestor Shachar A., et al., 2023, ApJ, 944, 78
Planck Collaboration et al., 2020, A&A, 641, A6
Riess A. G., et al., 2022, ApJ, 934, L7
Tacconi L. J., et al., 2018, ApJ, 853, 179
Zimmerman C. P., 2026, The galactic acceleration scale and the cosmological constant: the coefficient on SPARC, and what it takes to measure its redshift evolution, Zenodo (preprint of the long version), doi:10.5281/zenodo.23192441

This paper has been typeset from a TEX/LATEX file prepared by the author.